CHAPTER 6

The logarithm function

S44, S47, S49, S50

> **Key point · Objectives.** Use the properties of ln; solve equations and thresholds; differentiate and study logarithmic functions.

Course notes

Definition and properties

The exponential is continuous and strictly increasing on $\mathbb{R}$, and its range is $]0,+\infty[$. For every $x>0$, there is a unique real number y such that $e^y=x$: we denote it by $\ln x$.

$$\ln(e^y)=y \quad (y\in\mathbb{R}), \qquad e^{\ln x}=x \quad (x>0), \qquad \ln 1=0, \quad \ln e=1.$$

For $a,b>0$ and $n\in\mathbb{Z}$,

$$\ln(ab)=\ln a+\ln b, \quad \ln\left(\frac{a}{b}\right)=\ln a-\ln b, \quad \ln(a^n)=n\ln a, \quad \ln\sqrt{a}=\tfrac{1}{2}\ln a.$$

Proof of the product rule. $e^{\ln a+\ln b}=e^{\ln a}e^{\ln b}=ab$; the uniqueness of the preimage gives the formula. There is no analogous formula for $\ln(a+b)$.

Derivative, variations and limits

We accept that ln is differentiable on $]0,+\infty[$. Differentiating $e^{\ln x}=x$ gives $e^{\ln x}(\ln x)'=1$; hence $(\ln x)'=\frac{1}{x}>0$. ln is strictly increasing and concave, because its second derivative is $-\frac{1}{x^2}$.

$$\lim_{x\to 0^+}\ln x=-\infty, \qquad \lim_{x\to+\infty}\ln x=+\infty.$$

Its curve is the reflection of that of the exponential in the line $y=x$. The line $x=0$ is a vertical asymptote. Its tangent at 1 gives $\ln x\le x-1$ for $x>0$.

> **Key point · Composition and comparison of growth rates.** If $u>0$ is differentiable, $(\ln u)'=\frac{u'}{u}$. For every integer $m\ge 1$,
>
> $$\lim_{x\to+\infty}\frac{\ln x}{x^m}=0, \qquad \lim_{x\to 0^+}x^m\ln x=0.$$

Proof at 0. Set $t=-\ln x$; as $x\to 0^+$, $t\to+\infty$ and $x\ln x=-te^{-t}\to 0$ by comparison of growth rates. For $m\ge 1$, the same reasoning gives $-te^{-mt}\to 0$. At $+\infty$, set $t=\ln x$ to write $\frac{\ln x}{x^m}=\frac{t}{e^{mt}}\to 0$.

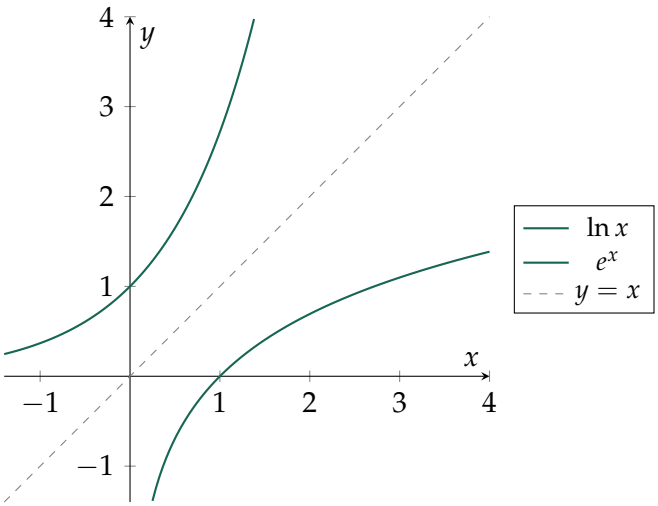
Equations and thresholds

For $A,B>0$, $\ln A=\ln B\Leftrightarrow A=B$ and $\ln A\le\ln B\Leftrightarrow A\le B$. Always start with the domain.

Example. $\ln(x-1)+\ln(x+1)=\ln 8$ requires $x>1$. The equation becomes $x^2-1=8$, so $x=3$ only.

Threshold. $0.8^n \leq 0.1$ is equivalent to $n \ln 0.8 \leq \ln 0.1$, then $n \geq \frac{\ln 0.1}{\ln 0.8} \simeq 10.319$. The first integer is 11.

Example of a study. For $f(x) = \frac{\ln x}{x}$, $f' = \frac{1-\ln x}{x^2}$; f increases on $]0, e]$, decreases on $[e, +\infty[$, and its maximum is $\frac{1}{e}$. Its limit at 0^+ is $-\infty$, its limit at $+\infty$ is 0.



Research activity

Controlling a calculation error — 30 min.

A calculation method produces an initial error $e_0 = 5$. At each step, the error is multiplied by 0.7. Denote by e_n the error after n steps, for $n \in \mathbb{N}$.

1. Express e_n in terms of n.
2. Using the logarithm, determine the smallest integer n such that $e_n \leq 10^{-4}$.
3. Using a calculator, calculate the error at the value found and at the previous value to check that it is minimal.
4. We now require an error of at most 10^{-6}. Determine the new minimum number of steps. Does dividing the tolerance by 100 multiply the number of steps by 100?

Progressive exercises

E1 · Practice. After determining the set on which it is defined, solve the equation $\ln(2x-1) = \ln(x+3)$.

E2 · Practice. Determine the real numbers x for which $\ln x + \ln(x-2)$ is defined, then solve $\ln x + \ln(x-2) = \ln 3$.

E3 · Practice. Consider $f(x) = \ln(x^2+1)$ and $g(x) = x\ln x$. State their domains of definition and calculate their derivatives.

E4 · Practice. Determine $\lim\limits_{x\to+\infty} \dfrac{\ln x}{\sqrt{x}}$ and $\lim\limits_{x\to 0^+} x^2 \ln x$. Justify the comparison-of-growth-rates results used.

E5 · Practice. By studying a function, prove that $\ln x \leq x-1$ for every real number $x > 0$. Do not use the concavity of the logarithm directly.

Textbook or past-paper work

Barbazo, p. 196, exercise 36. At home after S44: do the three functions. Write down the conditions of definition before solving the inequalities; justify the open endpoints of the intervals obtained.

Homework

Homework 6 — Study of a logarithmic function — 55 min.

Consider the function f defined on $(0, +\infty)$ by $f(x) = \frac{1+\ln x}{x}$. Its graph is denoted $\mathcal{C}_f$.

1. Determine the limits of f at the endpoints of its domain, study its sign, then draw up its table of variations.
2. Study the convexity of f and determine the coordinates of the point of inflection of $\mathcal{C}_f$.
3. Prove that the equation $f(x) = \frac{1}{2}$ has exactly two solutions. Give for each a justified bracketing using calculated values.
4. Using the maximum of f, prove that $1 + \ln x \leq x$ for all $x > 0$. State the case of equality.

Teaching notes and solutions

> **Teacher note · Teacher notes.** S44 is placed before the first mock exam, but the logarithm is not part of its announced scope. Stress the conditions $a > 0, b > 0$ before combining logarithms. The differentiability of ln is accepted; the differentiation formula is proved. For a threshold with $0 < q < 1$, the negative sign of $\ln q$ reverses the inequality. Have students check the candidate value and the previous one.

Solutions to the activity

1. Each step multiplies the error by 0.7: $e_n = 5(0.7)^n$ for $n \geq 0$.

2. $e_n \leq 10^{-4}$ is equivalent to $(0.7)^n \leq 2 \times 10^{-5}$. Since ln is increasing, $n \ln 0.7 \leq \ln(2 \times 10^{-5})$. As $\ln 0.7 < 0$, we obtain
$$n \geq \frac{\ln(2 \times 10^{-5})}{\ln 0.7} \simeq 30.33512.$$
The first suitable integer is 31.

3. $e_{30} \simeq 1.1270 \times 10^{-4} > 10^{-4}$ and $e_{31} \simeq 7.8888 \times 10^{-5} < 10^{-4}$. The sequence of errors is decreasing, so this check confirms minimality.

4. For the target 10^{-6}, the same calculation gives $n \geq \frac{\ln(2\times10^{-7})}{\ln 0.7} \simeq 43.2465$, so $n = 44$. The real threshold increases by $\frac{\ln 100}{|\ln 0.7|} \simeq 12.91$, then we round up to the next integer. The number of steps is not multiplied by 100.

E1. We need $2x - 1 > 0$ and $x + 3 > 0$, so $x > \frac{1}{2}$. On this domain, the injectivity of ln gives $2x - 1 = x + 3$, that is $x = 4$. This value belongs to the domain; the solution set is $\{4\}$.

E2. We need $x > 2$. The product rule gives $\ln(x(x-2)) = \ln 3$, so $x^2 - 2x - 3 = 0$, that is $(x-3)(x+1) = 0$. Only $x = 3$ satisfies $x > 2$: the other root -1 is excluded.

E3. Since $x^2 + 1 > 0$, the first function is differentiable on $\mathbb{R}$ and its derivative is $\frac{2x}{x^2+1}$. On $]0, +\infty[$, the product rule gives $(x \ln x)' = \ln x + x(\frac{1}{x}) = 1 + \ln x$.

E4. For the first limit, set $t = \sqrt{x} \to +\infty$: $\frac{\ln x}{\sqrt{x}} = \frac{2 \ln t}{t} \to 0$. For the second, set $t = -\ln x \to +\infty$: $x^2 \ln x = -te^{-2t} \to 0$. Both limits use comparison of growth rates.

E5. For $x > 0$, set $g(x) = x - 1 - \ln x$. We have $g'(x) = \frac{x-1}{x}$, negative before 1, positive after. Its minimum is $g(1) = 0$, so $g(x) \geq 0$, which is equivalent to $\ln x \leq x - 1$.

Solutions to Homework 6

1. At 0^+, the numerator tends to $-\infty$ and the denominator to 0^+: $f \to -\infty$. At $+\infty$, $f = \frac{1}{x} + \frac{\ln x}{x} \to 0$. The sign is that of $1 + \ln x$: negative on $]0, e^{-1}[$, zero at e^{-1}, positive afterwards. By the quotient rule,
$$f'(x) = \frac{1 - (1 + \ln x)}{x^2} = -\frac{\ln x}{x^2}.$$
Thus f strictly increases up to 1, then strictly decreases; its maximum is $f(1) = 1$.

2. Differentiating $-(\ln x)x^{-2}$, we obtain $f''(x) = \frac{2\ln x - 1}{x^3}$. The numerator changes sign, from negative to positive, at $x = \sqrt{e}$. So f is concave on $]0, \sqrt{e}]$, convex on $[\sqrt{e}, +\infty[$ and has the point of inflection $(\sqrt{e}, \frac{3}{2\sqrt{e}})$.

3. On $]0, 1]$, continuity, strict increase, limit $-\infty$ and final value 1 give exactly one preimage of $\frac{1}{2}$. On $[1, +\infty[$, strict decrease from 1 towards 0 gives exactly one second preimage. The first is in $]e^{-1}, \frac{1}{2}[$, because $f(e^{-1}) = 0$ and $f(\frac{1}{2}) = 2(1 - \ln 2) \simeq 0.6137 > \frac{1}{2}$. The second is in $]5, 6[$, because $f(5) \simeq 0.5219 > \frac{1}{2} > f(6) \simeq 0.4653$.

4. The maximum gives $\frac{1+\ln x}{x} \leq 1$. Since $x > 0$, we can multiply without changing the direction: $1 + \ln x \leq x$. Equality corresponds to the unique maximum of f, so to $x = 1$.

Solutions to the textbook or past-paper work

The logarithm requires a strictly positive argument.

1. $3x - 7 > 0 \iff x > \frac{7}{3}$, so $D_f = (\frac{7}{3}, +\infty)$.

2. $-x^2 + 4x - 3 = -(x-1)(x-3) > 0$ exactly when $1 < x < 3$. Thus $D_g = (1, 3)$.

3. We need simultaneously $x > 0$ and $2 - x > 0$; their intersection gives $D_h = (0, 2)$. The coefficient -3 placed in front of the second logarithm does not change its condition of definition.